

**INTEGRATION OF DEEP LEARNING
IN AUTOMATIC MUSIC GENERATION,
AIMING AT PRESERVING AND DEVELOPING
DON CA TAI TU**

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1. Introduction

The pervasive influence of contemporary music genres has significantly diminished public interest in Don ca tai tu – a traditional Southern Vietnamese musical art form recognized by UNESCO on December 5th, 2013, as an Intangible Cultural Heritage of Humanity. With the rapid advancement of digital technologies, numerous solutions have been implemented to broaden its accessibility and facilitate global exposure, cultural exchange, and audience engagement via social media platforms such as YouTube, Facebook, Spotify, etc. However, these initiatives have only succeeded in attracting a limited audience, while the majority, especially younger generations, remains largely uninterested in this traditional musical genre, revealing a persistent generational gap in appreciation for this heritage art form. This phenomenon underscores an urgent imperative: the integration of cutting-edge technological interventions is critical not only for preserving Don ca tai tu but also for reinvigorating its appeal across diverse populations, particularly among youth.

In recent years, Artificial Intelligence (AI) has made remarkable progress in the field of automatic music composition [Copet et al., 2023; Evans et al., 2024]. With the advent of Deep Learning, its potential is believed to surpass that of traditional approaches in generating music. However, applying Deep Learning to compose Don ca tai tu presents several key challenges: (1) *Data Scarcity*: the lack of large-scale, high-quality paired datasets of text description and music audio hinders model performance. (2) *Long-form Structure*: Don ca tai tu compositions often span several minutes, posing difficulties for models to maintain coherence and fidelity over extended durations. (3) *Signal Complexity*: Unlike speech or environmental sounds, music is inherently polyphonic and demands precise, harmonious coordination among multiple instrumental components [Yuan et al., 2025]. (4) *Domain Complexity*: Human listeners are particularly attuned to musical dissonance, leaving little room for error in generated compositions [Copet et al., 2023].

Furthermore, enabling user-driven control and personalization of the music generation process through natural language prompts is pivotal for democratizing composition, allowing individuals without specialized expertise to engage creatively with Don ca tai tu music. This approach not only facilitates preservation but also modernizes the art form, fostering a connection between younger generations and traditional values through advanced Deep Learning technologies. By addressing these challenges, our work pioneers a novel framework for cultural preservation and revitalization, ensuring that Don ca tai tu resonates with contemporary contexts and engages diverse audiences.

1.1. Research Objective

Develop a unified Deep Learning framework to generate high-fidelity, long-form compositions infused with the stylistic essence of Don ca tai tu based on text inputs.

1.2. Research Questions

- What characteristics define a clean, high-quality dataset capable of effectively training Deep Learning models for optimal performance?
- Which Tokenizer model can compress music audio at ultra-low bitrates while preserving critical semantic and acoustic information?
- Which Deep Learning model supports the maintenance of long-term musical structure in generated compositions?
- Which Deep Learning model facilitates the creation of music with high fidelity?
- Which Tokenizer model efficiently extracts semantic information from text input to enable user-driven personalization of generated music?
- What technique enhances the coherence and creativity of Deep Learning models in text-to-music generation?

1.3. Hypotheses

- Collecting recordings from multiple online sources, experienced musicians, and cultural experts and meticulously filtering and preprocessing can ensure a clean, high-quality dataset for model training.
- MuCodec [Xu et al., 2024] can compress music audio at ultra-low bitrates while preserving critical semantic and acoustic features.
- Autoregressive Transformer [Vaswani et al., 2017] plays an important role in maintaining long-form musical structure for generated compositions.
- Rectified Flow Transformer [Esser et al., 2024] can produce fine-grained acoustic representations, which facilitates the creation of music with high-fidelity audio.
- Qwen 2.5 Tokenizer [Yang et al., 2024] can efficiently extracts semantic information from text input without the need for additional training.
- Chain-of-Thought prompting technique [Wei et al., 2022] tailored for music generation can enhance the coherence and creativity of resulting compositions.

2. Related Work and Preliminaries

2.1. Current Datasets

Over the past decade, numerous music datasets have been developed to advance the application of Deep Learning in music generation. Prominent datasets such as Free Music Archive [Defferrard et al., 2016], MTG-Jamendo [Bogdanov et al., 2019], and Music4ALL [Santana et al., 2020] provide extensive collections of high-quality, genre-diverse audio files, enabling researches to harness the full potential of Deep Learning models. However, these datasets lack accompanying textual descriptions

for the audio files, limiting the ability to personalize generated compositions based on user preferences.

Name	Year	Files	Text captions
Free Music Archive	2016	106,574 samples, 8,200 hours	No
MTG-Jamendo	2019	55,701 samples, 3,800 hours	No
Music4ALL	2020	109,269 samples, 911 hours	No
MusicCaps	2023	5,521 samples, 55,210 hours	Yes
LP-MusicCaps-MSD	2023	513,977 samples, 15,419,310 hours	Yes
Vietnamese Traditional Music Dataset	2023	2,500 samples, 21 hours	No

Table 1: Current open-source notable music datasets

To address this limitation, MusicCaps [Agostinelli et al., 2022] and LP-MusicCaps-MSD [Doh et al., 2023] pair each audio file with corresponding textual captions, facilitating the training of text-to-music generation models. Despite these advances, none of the existing datasets include recordings of Don ca tai tu, posing significant challenges for applying Deep Learning models tailored to this traditional art form. In 2023, Ho Manh Thang published the Vietnamese Traditional Music Dataset on Kaggle, offering a valuable resource for integrating Deep Learning for Vietnamese traditional music research. However, this dataset exhibits several limitations: (1) The audio files retain vocal elements, introducing noise that compromises the quality of generated compositions. (2) The original recordings are splitted into fixed 30-second clips, disrupting the inherent musical structure and continuity essential for coherent outputs. (3) The files are not accompanied by corresponding textual annotations. (4) The dataset excludes Don ca tai tu recordings, hampering model development specific to this traditional music genre.

2.2. Audio Tokenizers

Recent advancements in audio generation have positioned audio codecs based on the VQ-GAN framework as a prominent paradigm for discrete audio representations. Notably, Encodec [Défossez et al., 2022] introduces a versatile architecture adaptable to diverse audio types by integrating Residual Vector Quantization (RVQ) and Generative Adversarial Network (GAN) to optimize audio compression and reconstruction. The model further enhances compression efficiency through a multi-scale discriminator and a balanced loss strategy, augmented by a language model to improve audio reconstruction from discrete representations. HiFiCodec [Yang et al., 2023] employs Group-Residual Vector Quantization (GRVQ) to minimize codebook requirements while maintaining high-fidelity audio reconstruction. WavTokenizer [Ji et al., 2024] functions as an efficient audio tokenizer with only 1

Codec Name	Reconstruction Quality	LM Converge	Invalid Prob	Specialized for music	Sample rate (KHz)	Tokens rate
Encodec	Good	No	All	No	24	75/sec
HiFiCodec	Good	No	All	No	24	75, 100/sec
WavTokenizer	Good	No	All	No	24	40, 75/sec
SemantiCodec	Fair	Yes	High	No	16	25, 50, 100/sec
X-Codec	Fair	Yes	Low	No	16	50/sec
MuCodec	Good	Yes	Low	Yes	32	25/sec

Table 2: Comparison of different codec types

codebook by designing a broader Vector Quantization (VQ) space, extending contextual windows, incorporating attention mechanisms in the decoder, and introducing a multi-scale discriminator along with an inverse Fourier Transform in the vocoder. However, these codecs primarily focus on acoustic reconstruction and higher compression rates, overlooking their potential as tokenizers for audio LLMs.

To address this challenge, recent studies have proposed specialized tokenizers tailored for audio LLMs. SemantiCodec [Liu et al., 2024] utilizes a pre-trained AudioMAE [Huang et al., 2022] to generate distinct semantic and acoustic tokens from mel-spectrograms. However, since AudioMAE operates on 2D time-frequency mel-spectrograms, LLMs must process both dimensions concurrently, necessitating significant modifications to existing LLM architectures. In contrast, X-Codec introduces a mechanism to enhance semantic consistency across all tokens, substantially improving the performance of audio LLMs without requiring structural alterations to the model architecture. However, these models are not specifically designed for music-related tasks. Compared to speech and environmental audio, music signals are highly complex, exhibiting both short-term melodic coherence and long-term structural consistency. To tackle this problem, MuCodec [Xu et al., 2024] is proposed to specifically target music compression and reconstruction tasks at ultra-low bitrates by employing MuEncoder as the feature extractor and Diffusion Transformer (DiT) [Peebles et al., 2022] along with flow-matching-based method for fine-grained music modeling.

2.3. Music Generation Models

Text-to-music generation models can be categorized into 2 primary paradigms: **Autoregressive (AR)** and **Non-Autoregressive (NAR)** models.

AR models draw inspiration from the successful work done in natural language processing, modeling sequences by predicting each token based on preceding ones. Recent prominent models utilize multi-stream quantized autoencoders [Défossez et al., 2022] to transform audio into K parallel discrete token streams to extract critical

features from the audio while reducing dimensionality to facilitate fast training and inference of DL model. Subsequently, an AR Transformer [Vaswani et al., 2017] is employed to model these discrete audio tokens. MusicLM [Agostinelli et al., 2023] and MusicGen [Copet et al., 2023] are built on this principle. MusicLM adopts a multi-stage processing strategy with separate models to predict K streams, whereas MusicGen processes these streams in parallel using a delayed codebook pattern. While these models demonstrate the ability to generate convincing short audio segments, they often struggle with maintaining coherence over extended durations, leading to repetitive or meandering outputs.

NAR models are latent diffusion models operating in the latent space of a continuous variational autoencoder. Stable Audio Open employs latent diffusion to iteratively denoise audio representations, achieving impressive perceptual quality. Similarly, AudioLDM2 [Liu et al., 2023] leverages latent diffusion conditioned on text, offering an efficient pathway to generating complex audio. MeLoDy [Lam et al., 2023] implements a language model to process coarse semantic tokens and a dual path diffusion model to model acoustic representations. However, these methods often sacrifice computational efficiency and in some cases hard to maintain long-term structural coherence in the generated music. Despite their strengths in synthesizing high-fidelity outputs, these models generally suffer from a lack of fine-grained control when working with long and intricate musical structures.

2.4. Chain-of-Thought Prompting

Wei et al. (2022) introduced the Chain-of-Thought (CoT) prompting paradigm, a novel approach that elicits a series of intermediate reasoning steps within Large Language Models (LLMs). Their empirical findings indicate that CoT significantly enhance the reasoning capabilities of these models, catalyzing a surge of subsequent studies [Zhang et al., 2023; Feng et al., 2023]. Notably, CoT has emerged as a cornerstone of cutting-edge industrial innovations, including OpenAI’s o1 [Jaech et al., 2024] and DeepSeek-R1 [Guo et al., 2025]. These advancements underscore CoT’s practical utility and transformative potential in refining language model performance for complex and real-world tasks.

Recent studies in the music audio domain have begun to explore the application of CoT prompting technique [Yuan et al., 2025; Lam et al., 2025]. Specifically, Yuan et al., [2025] introduced a CoT method that focuses on music segments (e.g., intro, verse, chorus, bridge, and outro) extracted by the all-in-one model [Kim & Nam, 2023] to address the long-term decay property issue. In contrast, Lam et al., [2025] offers a more nuanced and abstractive paradigm for music generation, emphasizing that reasoning within a latent space enables LLMs to perform flexible inference without the constraints of natural language.

3. Methodology

3.1. Overview

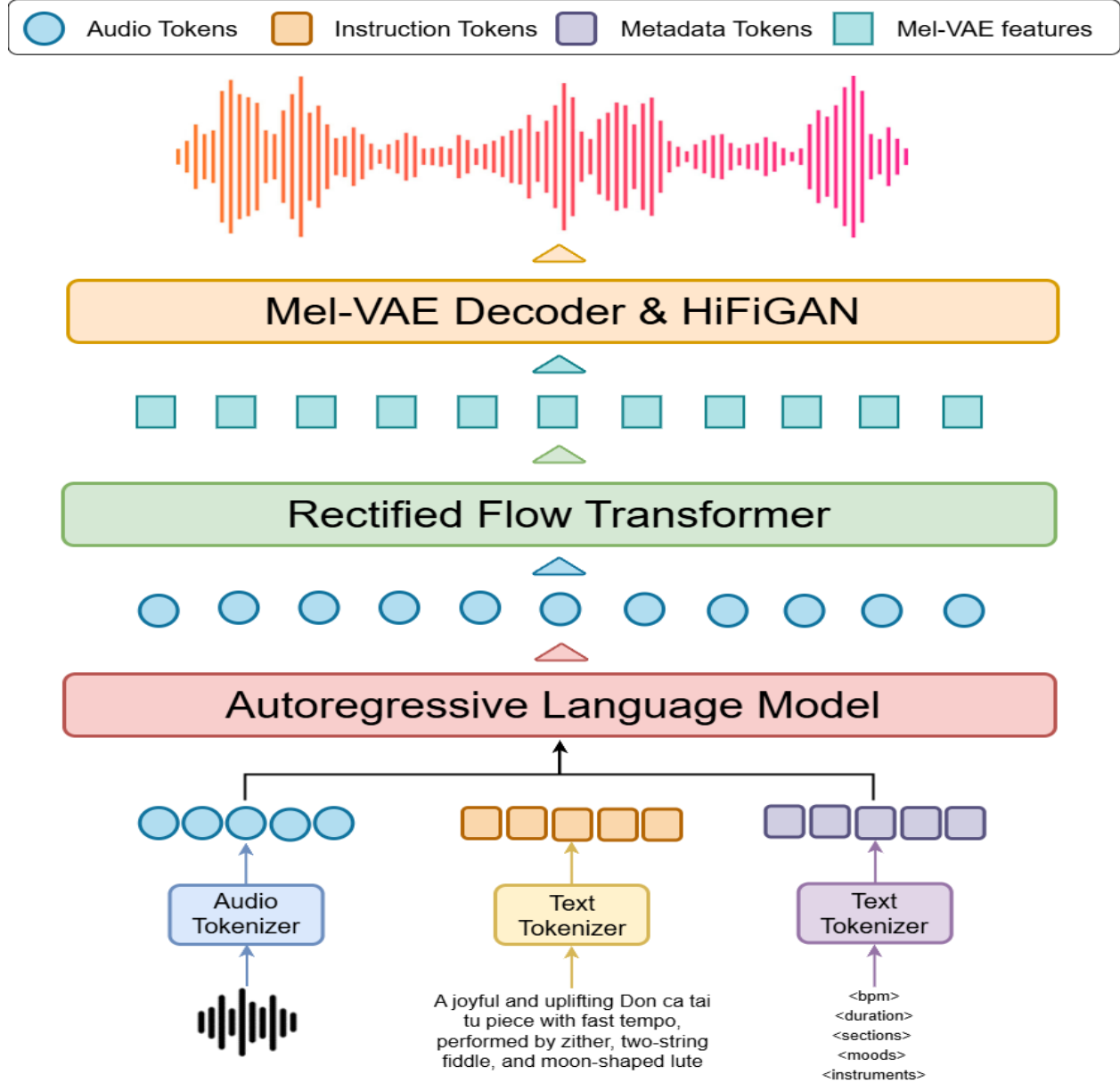


Figure 1: The overview of DcttGen

We introduced DcttGen, a unified Deep Learning framework to generate high-fidelity, long form compositions infused with the stylistic essence of Don ca tai tu. Specifically, DcttGen comprises four main components: an audio tokenizer, a text tokenizer, an autoregressive transformer, and a rectified flow transformer. The audio tokenizer converts waveforms into discrete tokens to capture essential musical features while compressing the sequence into a more compact representation. The autoregressive transformer [Vaswani et al., 2017] is trained on text tokens and semantic audio tokens, modeling text-to-music generation as an AR next-token

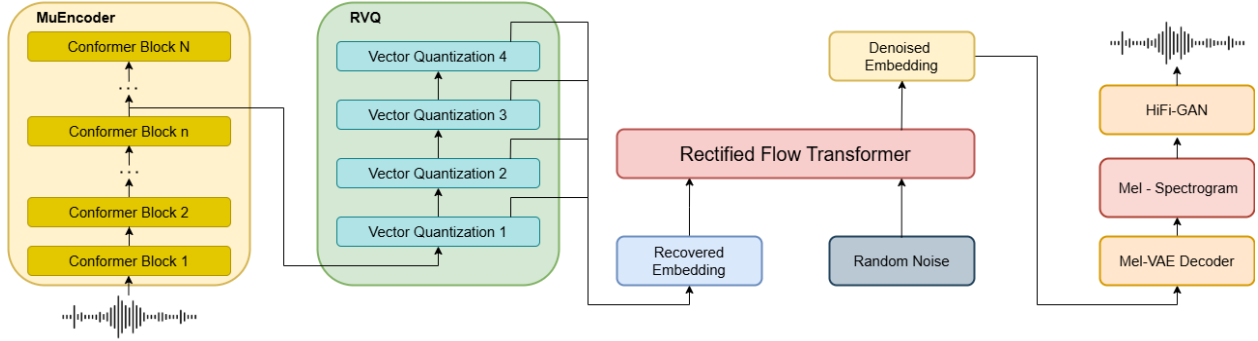


Figure 2: The framework of MuCodec [Xu et al., 2024]. We replaced the flow-matching model with Rectified Flow Transformer [Esser et al., 2024] to obtain high-resolution Mel-VAE features with few-step ODE sampling

prediction (NTP) tasks. The objective of rectified flow transformer [Esser et al., 2024] is to reconstruct discrete representations to obtain Mel-VAE features, facilitating high quality audio outputs. Subsequently, the pre-trained Mel-VAE decoder [Liu et al., 2024] restores these features into a Mel spectrogram. Finally, the generated music is obtained through a pre-trained HiFiGAN [Kong et al., 2020].

3.2. Audio and Text Tokenization

For text tokenization, we utilized Qwen 2.5 Tokenizer [Yang et al., 2024], which implements byte-level byte-pair encoding [BBPE] with a vocabulary of 151,643 regular tokens. Instructions, metadata, structure annotations, and structure segment boundary signals are represented with text format and tokenized with BBPE. Building upon the foundational work of Yuan et al., [2025], we expanded the tokenizer vocabulary to support additional special tokens, including: <EOD> represents the end of a document, <SOA> denotes the start of an audio segment, <EOA> signifies the end of an audio segment, <INST> indicates the start of an instruction, and <PLAN> presents the start of a metadata, which includes music elements aligned with the music audio (e.g., bpm, duration, moods, etc.)

For the audio part, we experimented with several open-source music and universal audio codecs. Based on the comparison table in section 2.2, we employed MuCodec [Xu et al., 2024] as an audio tokenizer in DcttGen. It compresses 32KHz audio into discrete tokens at a 25Hz token rate. These tokens effectively capture both acoustic and semantic features of music, while significantly reducing sequence length to facilitate more efficient training of autoregressive Transformer models. The efficacy of MuCodec is driven by its MuEncoder architecture, comprising 13 stacked Conformer blocks and 4 RVQ layers, each with a codebook size of 10,000. Additionally, the framework synergistically combines flow matching for fine-

grained reconstruction, a pre-trained Mel-VAE decoder for Mel spectrogram generation, and a pre-trained HiFiGAN vocoder to ensure high-fidelity waveform synthesis. MuCodec is trained on a large-scale music dataset, with each training example standardized to a fixed duration of 35.84 seconds and a minimum of 32KHz sampling rate, ensuring robust representation of complex musical structures.

3.3. Autoregressive Transformer

3.3.1. Model Architecture

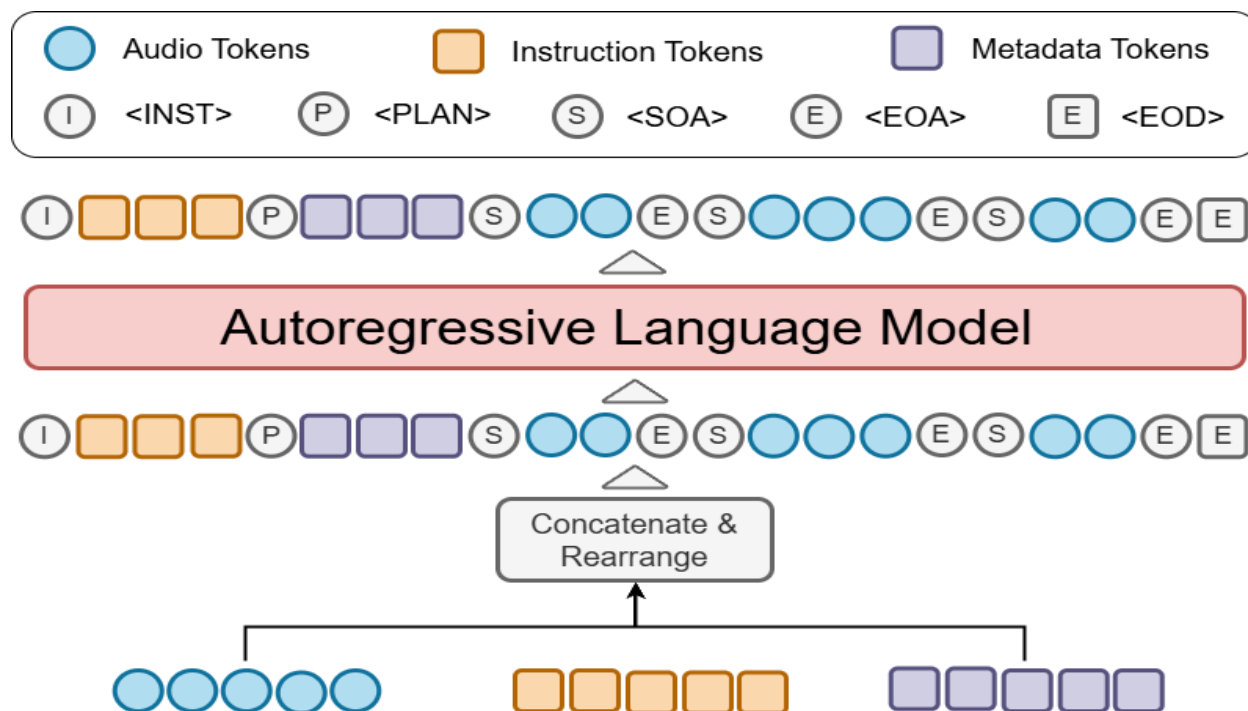


Figure 3: The Autoregressive Language Model of DcttGen

The core of DcttGen is an autoregressive transformer [Vaswani et al., 2017], with the backbone large language model of Qwen 2.5 model [Yang et al., 2024]. The model learns to predict the next audio token in the sequence based on the preceding tokens and conditional inputs, thereby generating extended sequences of semantic audio tokens with long-range temporal coherence and structural integrity.

The selection of Qwen 2.5 as the backbone language model is motivated by its exceptional performance, flexibility, and scalability, making it particularly suitable for text-to-music generation tasks that demand the capture of long-range dependencies to produce cohesive, well-structured compositions from text prompts. Specifically, The architecture of Qwen 2.5 incorporates several cutting-edge techniques to address music generation challenges: Group Query Attention [Ainslie et al., 2023] to optimize key-value cache efficiency, reducing computational overhead while maintaining high performance, SwiGLU activation function [Dauphin et al., 2017] for non-linear activation, elevating the model’s capacity to

manage complex data distributions, Rotary Position Embedding [Su et al., 2024] for encoding position information and enabling robust extrapolation to extended sequence lengths, QKV bias [Su, 2023] in the attention mechanism to better capture intricate dependencies within the data, and RMSNorm [Jiang et al., 2023] with pre-normalization to facilitate stable and efficient optimization during training.

3.3.2. Music Chain-of-Thought

Challenges of generating a full song of Don ca tai tu: Don ca tai tu pieces often span several minutes, requiring deep learning models to handle minutes-long context. Although some proprietary systems like Suno.ai and Udio achieved this, their methodologies remain undisclosed. To tackle this problem, Yuan et al., [2025] proposed a solution that leverages the inherent structural priors of music. Their motivation stemmed from researches indicating that songs are typically composed of distinct segments, such as intro, verse, chorus, bridge, and outro. However, Don ca tai tu is inherently characterized by its high degree of improvisational expression, which often results in compositions lacking a well-structured framework. Specifically, upon consulting experience musicians and domain experts, we observed that unlike conventional modern musical forms, Don ca tai tu does not segment the body of a composition into distinct verses or choruses. Rather, performers engage in dynamic, collaborative improvisation, eschewing fixed predetermined architectures. This intrinsic characteristic poses significant challenges to the application of the all-in-one [Kim & Nam, 2023], such as that proposed by Yuan et al., [2025], to automatically segment songs into musical sections.

Music Chain-of-Thought: To address the aforementioned challenges, we redesigned the Chain-of-Thought prompting technique tailored for Don ca tai tu generation. Since the body of a composition cannot be split into verses or choruses, we retained this part and segmented the composition into 3 sections: intro, main, and outro. We set the maximum duration of intro and outro to 30 seconds, and the main part is 240 seconds. Special tokens were incorporated to indicate the start and end of audio and to separate the discrete audio tokens with instruction and metadata text tokens (see token arrangement in Figure 2).

We denote a single training example as a concatenation of several components. In our setup, a composition is constructed as:

$$\mathcal{X} = \text{Instruct} \odot \text{Metadata} \odot \text{Audio Segments} \odot \langle \text{EOD} \rangle$$

Where $\odot$ denotes the concatenation operation. “Instruct” is the text description corresponded to the music audio:

“<INST> A joyful and uplifting Don ca tai tu piece with fast tempo, performed by zither, two-string fiddle, and moon-shaped lute”

“Metadata” signifies the musical elements, which is the style control string. An example of Metadata is as follows:

“<PLAN> bpm: 80; duration: 150; Sections: [intro] 30, [main] 90, [outro] 30 ; moods: uplifting, joyful; instruments: zither, two-string fiddle, moon-shaped lute”

Besides, each audio segment is structured as follow:

$$\mathcal{S} = [start_of_seg] \odot \tau \odot <SOA> \odot \psi \odot <EOA> \odot [end_of_seg]$$

Where $\tau \in \{[intro], [main], [outro]\}$ is a structure label and ψ is the sequence of audio tokens.

In summary, each training example begins with an instruction and metadata, followed by a series of annotated segments, and ends with the <EOD> tokens. Furthermore, we utilized the coarse-to-fine flattened RVQ prediction strategy [Lam et al., 2025]. Specifically, we arranged RVQ tokens in a flattened coarse-to-fine sequence for LM prediction, ensuring that coarser tokens are predicted before finer ones. As the codebooks represent different levels of granularity [Zeghidour et al., 2021], flattening these codebooks makes the generation of these tokens akin to structuring music from a broad to detailed perspective, thereby capturing both semantic and acoustic features of complex compositions in Don ca tai tu effectively. We adopted the standard next-token prediction (NTP) paradigm to model the token sequences. Let $x_{1:T}^{(0:3)} = (x_1^{(0)}, \dots, x_T^{(0)}, \dots, x_1^{(3)}, \dots, x_T^{(3)})$ represent the sequence of audio tokens, where each $x_t^{(m)}$ corresponds to the token at timestep t and belongs to codebook m . In the standard NTP framework, we factorized the joint probability of $x_{1:T}^{(0:3)}$ as:

$$p(x_{1:T}^{(0:3)}) = \prod_{t=1}^{T \times 4} p(x_t | x_{<t}, C)$$

Where $C = \{c_{inst}, c_{plan}\}$ denotes the instruction condition and metadata condition respectively. During inference, the model predicts the next token $\hat{x}_t$ which maximizes the conditional distribution:

$$\hat{x}_t = \operatorname{argmax} p(x_t | x_{<t}, C)$$

The training objective of AR language model can be formulated as:

$$\mathcal{L}_{AR} = - \sum_{t=1}^{T \times 4} \log p(x_t | x_{<t}, C)$$

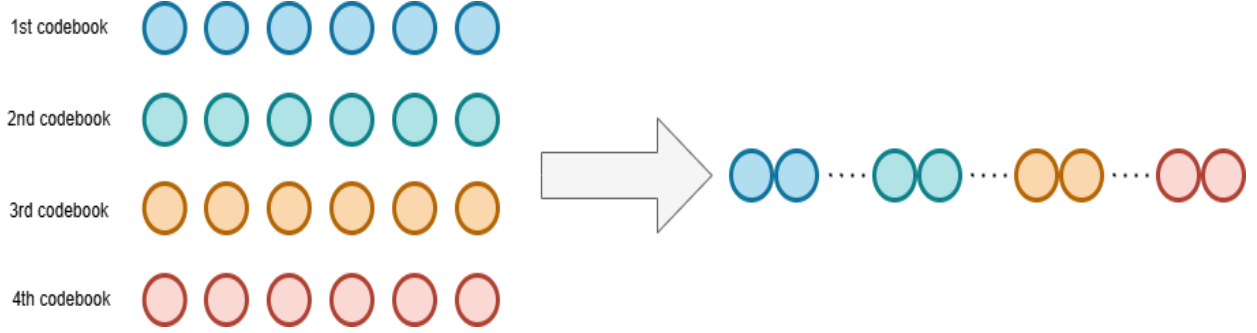


Figure 4: Coarse-to-fine RVQ prediction strategy

3.4. Rectified Flow Transformer

To achieve high-fidelity music reconstruction at ultra-low bitrates, MuCodec employs a flow-matching model to produce information-rich Mel-VAE features conditioned by the discretized MuEncoder representations. However, the primary issue with flow-matching model is their reliance on curved generative trajectories, which necessitate numerous integration steps during sampling, leading to increased computational overhead and slower inference times [Liu et al., 2022]. To tackle this problem, we replaced the flow-matching component with rectified flow transformer. Specifically, rectified flows [Liu et al., 2022; Lipman et al., 2023] define the forward process as straight paths between the noise distribution and data distribution:

$$z_t = (1 - t)x_0 + t\varepsilon$$

Where z_t is the latent audio representation, $t \in [0,1]$ is the time embedding, x_0 is a sample from the data distribution p_0 , and ε is random Gaussian noise. After redefining the flow trajectory, the model objective can be formulated as:

$$\mathcal{L} = \mathbb{E}_{t, p_t(z|\varepsilon, c), p(\varepsilon, c)} \|v_\theta(z, t; c) - u_t(z|\varepsilon, c)\|_2^2$$

Where c is the conditioned discretized MuEncoder representation, p_t is the probability path transitioning from p_0 to $p_1 = \mathcal{N}(0, 1)$, $p_t(z_t) = \mathbb{E}_{\varepsilon \sim \mathcal{N}(0, I)} p_t(z_t|\varepsilon)$ are the marginals align with both the data and noise distribution, $v_\theta(\cdot)$ is the vector field parameterized by the weights θ of a neural network, and $u_t(z) = \mathbb{E}_{\varepsilon \sim \mathcal{N}(0, I)} \left[u_t(z_t|\varepsilon) \frac{p_t(z_t|\varepsilon)}{p_t(z_t)} \right]$ is a marginal vector field that generates the marginal probability paths p_t using the conditional vector fields $u_t(\cdot|\varepsilon)$. The model uses $\mathcal{L}$ which then corresponds to $w_t^{RF} = \frac{t}{1-t}$ following Kingma & Gao [2023] form. The network output directly parameterizes the vector field v_θ .

In summary, the rectified flow transformer was employed to reduce the number of integration steps by enforcing straight-line trajectories, thereby enabling fast training

and inference while maintaining high-fidelity audio reconstruction from low sampling rate, semantic, discrete tokens generated from the AR language model.

3.5. Data Preprocessing Pipeline

We assembled Don ca tai tu recordings over extended durations from multiple online sources, experienced musicians, and domain experts. The audio data was resampled into 32KHz sampling rate and downmixed to mono. We employed Fr chet Audio Distance (FAD) [Kilgour et al., 2019] to evaluate the quality of the music audio files by extracting 3 distinct embeddings from VGGish [Hershey et al., 2017], CLAP [Wu et al., 2023], and EnCodec [D fossez et al., 2022] and then calculating the mean value of these embeddings to achieve the FAD score for each file. Samples with low FAD scores were discarded, ensuring only high-quality recordings are kept for model training.

After the data filtering process, we leveraged librosa to extract the bpm, duration, and sampling rate information for each music audio file and Essentia library to capture the moods conveyed by the music. In Don ca tai tu, six traditional instruments are commonly used are the moon-shaped lute, Vietnamese two-string fiddle, Vietnamese zither, monochord, bamboo flute, and gong ban. To collect information regarding the presence of these instruments in musical pieces, we conducted a listening process and annotated the instrument appearances manually. Finally, we used ChatGPT 4o API combined with all extracted musical elements to produce the text caption corresponding to each music audio file. After preprocessing, the dataset contains about 28,000 clips, totaling around 2,000 hours of audio.

3.6. Training and Inference Strategies

The training procedure of DcctGen includes training the autoregressive language model and rectified flow transformer model. We used 8 NVIDIA H100 GPUs with 80GB memory for both training and experimenting.

The autoregressive language model undergoes a two-phase training process: Pre-training and Fine-tuning. In the first stage, the model was trained with semantic discrete tokens extracted via MuCodec. In the second stage, the model was trained with text descriptions and metadata alongside the corresponding audio tokens.

The rectified flow transformer model was trained using paired coarse semantic MuEncoder representations and fine-grained Mel-VAE features to facilitate high-fidelity reconstruction for music audio.

All models were trained using AdamW optimizer with a base learning rate of 3×10^{-4} , weight decay 0.1, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-9}$. The learning rate follows a linear warm-up and cosine annealing schedule with a maximum learning rate of 3×10^{-5} . In the AR language model decoding process, only vocabulary tokens within the audio range are permitted until the <EOA> token is predicted.

4. Experiments

4.1. Experimental Setup

To evaluate and compare the performance of our proposed approach with other methods in Don ca tai tu generation, we splitted our dataset into training and validating set with the ratio of 9.8/0.2. We used 4 foundational baselines including MusicLM [Agostinelli et al., 2023], MusicGen [Copet et al., 2023], Stable Audio Open [Evans et al., 2024], and AudioLDM 2 [Liu et al., 2023] for the evaluation part. For MusicGen, Stable Audio Open, and AudioLDM 2, we leveraged the open-source implementation and checkpoints and trained with our Don ca tai tu dataset. For MusicLM, we utilized the implementation of lucidrain on GitHub to train the model. Detailed results are presented in section 4.3.

4.2. Evaluation Metrics

We evaluated the proposed framework with 3 objective metrics: the Fréchet Audio Distance (FAD), the Kullback-Leibler Divergen (KLD), and the CLAP Score.

FAD_{openl3} : The Fréchet Audio Distance (FAD) [Kilgour et al., 2019] evaluates the similarity between the statistics of a generated audio set and a reference audio set in a feature space. A low FAD_{openl3} score indicates that the generated audio is plausible and closely matches the reference.

KLD_{passt} : We used PaSST [Koutini et al., 2022], a state-of-the-art audio tagger trained on AudioSet [Gemmeke et al., 2017], to compute the Kullback–Leibler divergence (KLD) over the probabilities of the labels between the generated and the reference audio. It computes the KL divergence over the probabilities of the labels between the original and the generated music. The generated music is expected to share concepts similar to the reference music when the KLD is low.

$CLAP_{score}$: The CLAP score [Wu et al., 2023] is computed between the music description and the generated audio to quantify audio-text alignment, using the official pre-trained CLAP model.

4.3. Results

Model	$FAD_{openl3}\downarrow$	$KLD_{passt}\downarrow$	$CLAP_{score}\uparrow$
MusicLM	4.023	0.822	0.305
MusicGen-Medium	3.816	0.756	0.318
AudioLDM2-Music	4.041	0.793	0.346
Stable Audio Open	1.378	0.519	0.410
DccttGen	1.291	0.472	0.394

Table 3: The objective evaluation of DccttGen with MusicLM, MusicGen-Medium, AudioLDM2-Music, and Stable Audio Open

We reported FAD and KLD to evaluate how well generated audio matches the target distribution. Result presents that our proposed framework achieves the lowest FAD score, outperforming prior transformer-based and diffusion-based baselines such as MusicGen-Medium (3.816) and Stable Audio Open (1.378). Furthermore, DccttGen also achieves the best performance on KLD with 0.472 compared to others. This result demonstrates the model’s ability to generate high-fidelity outputs and

distribution matching capabilities. Although Stable Audio Open surpasses our framework in CLAP score, our approach maintains a highly competitive alignment between music and the conditioning text. These results highlight the effectiveness of our model in generating high-fidelity, semantically coherent, and text-aligned music. These findings underscore the competitive performance of our architecture and design choices, including the utilization of MuCodec, autoregressive language model, rectified flow transformer, and the redesigned Chain-of-Thought technique.

5. Conclusion and Future Work

We introduced DcttGen, a unified framework designed for Don ca tai tu generation based on natural language prompts. By combining a specialized dataset for Don ca tai tu, MuCodec tokenizer, Qwen 2.5 as the backbone autoregressive language model, rectified flow transformer, and music chain-of-thought technique, DcttGen can generate high-fidelity, long-form coherent compositions with detailed musical structure. Experimental results show that DcttGen matches or even surpasses several recent methods on various objective evaluation metrics. Through this work, we presented a novel approach to the preservation and revitalization of Don ca tai tu.

DcttGen’s approach can be extended by incorporating vocals into the composition and enabling users to control the generated musical pieces with lyrical content. Moreover, we planned to employ the in-context learning (ICL) framework to further enhance user interaction with the system by allowing them to adjust the resulting compositions based on a reference audio track (instrumental, vocal, or both). Furthermore, the creation of an innovative module designed to blend the traditional Vietnamese art form of Don ca tai tu with modern musical genres holds the potential to captivate a broader contemporary audience, especially younger generations. This approach not only increases the appeal and accessibility of Don ca tai tu but also significantly contributes to the preservation, promotion, and development of this intangible cultural heritage, ensuring its vitality in the modern era.

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